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Insertion device for a biosensor and insertion method thereof

Abstract

An insertion device includes a cover body, an insertion module. The insertion module is disposed in the cover body, and includes a main body, an insertion seat, a first elastic member, a retraction seat and a second elastic member. When the cover body is depressed, the insertion seat is driven by the first elastic member to perform an automatic-insertion operation and to collapse a limiting structure between the insertion seat and the cover body. A limiting structure between the insertion seat and the retraction seat collapses upon the collapse of the limiting structure between the insertion seat and the cover body, so that the retraction seat is driven by the second elastic member to perform an automatic-retraction operation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority of U.S. Provisional Patent Application No. 62/882,140, filed on Aug. 2, 2019, and priority of Taiwanese Invention Patent Application No. 109100967, filed on Jan. 10, 2020.

FIELD

(1) The disclosure relates to an insertion device and an insertion method thereof, and more particularly to an insertion device for a biosensor and an insertion method thereof.

BACKGROUND

(2) A conventional insertion device disclosed in U.S. patent Ser. No. 10/413,183 is for inserting a biosensor into a host, and includes a plunger, and a piercing member to which the sensor assembly is mounted.

(3) The piercing member is inserted into the host upon depression of the plunger, but is not separated from the host until the plunger is released. Accordingly, the conventional insertion device

is operated via a two-action operation. When an operator fails to promptly release the plunger, the piercing member remains to be inserted into the host and may cause discomfort of the host.

SUMMARY

(4) Therefore, an object of the disclosure is to provide an insertion device that can alleviate the drawback of the prior art.

(5) According to one aspect of the disclosure, the insertion device includes a cover body, an insertion module. The cover body has an accommodating space. The insertion module is disposed in the accommodating space, and includes a main body, an insertion seat, a first pre-compressed elastic member, a retraction seat, a second pre-compressed elastic member, a base and a sensor assembly. The main body is coupled to the cover body, and cooperates with the cover body to define a displacement space therebetween. The insertion seat is removably positioned in the cover body, and is able to move in the displacement space between the main body and the cover body. The first pre-compressed elastic member has two opposite ends that respectively abut against the insertion seat and the cover body. The retraction seat is removably positioned relative to the insertion seat. The second elastic member has two opposite ends that respectively abut against the insertion seat and the retraction seat. The base is separably positioned relative to the main body. The sensor assembly is to be separably mounted to the base. When the cover body is depressed, the insertion seat is driven by a restoring force of the first elastic member to perform an automatic-insertion operation and to collapse a limiting structure between the insertion seat and the cover body. After the automatic-insertion operation is done, the cover body is positioned relative to the main body without rebounding, and a limiting structure between the insertion seat and the retraction seat collapses upon the collapse of the limiting structure between the insertion seat and the cover body, so that the retraction seat is driven by a restoring force of the second elastic member to perform an automatic-retraction operation.

(6) Another object of the disclosure is to provide an insertion method that can alleviate the drawback of the prior art.

(7) According to one aspect of the disclosure, the insertion method includes: an insertion device including a displacement space that is cooperatively defined by a cover body and a main body; an insertion module disposed in the displacement space, the insertion module being movable in the displacement space by compressed first and second elastic members therein so as to perform an automatic-insertion operation and an automatic-retraction operation; an insertion limiting structure disposed between the insertion module and the main body for removably positioning the insertion module at a pre-insertion position; and a retraction limiting structure disposed between the insertion module and the cover body, the insertion module being able to move from a post-insertion position to the pre-insertion position when the retraction limiting structure is collapsed. When the insertion device is kept being depressed, the insertion limiting structure is collapsed, and the insertion module is biased by a restoring force of the first elastic member to move downwardly to perform the automatic-insertion operation. The retraction limiting structure is collapsed immediately after the insertion module finishing the automatic-insertion operation, such that the insertion module is biased by a restoring force of the second elastic member to move upwardly to perform the automatic-retraction operation. The insertion module positions a sensor assembly thereof onto a base during the automatic-insertion operation. A time needed for performing the automatic-insertion operation and the automatic-retraction operation is no more than 100 milliseconds.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other features and advantages of the disclosure will become apparent in the following detailed

description of the embodiments with reference to the accompanying drawings, of which:

- (2) FIG. 1 is an exploded perspective view illustrating a first embodiment of the insertion device according to the disclosure;
- (3) FIG. 2 is another exploded perspective view illustrating the first embodiment;
- (4) FIG. 3 is a sectional view illustrating of first embodiment, illustrating a sensor of a container in a state to be implanted;
- (5) FIG. 4 is a sectional view taken along line IV-IV in FIG. 3;
- (6) FIG. 5 is a sectional view illustrating a lower casing of the first embodiment uncovering a base;
- (7) FIG. 6 is a sectional view taken along line VI-VI in FIG. 5;
- (8) FIG. 7 is a sectional view illustrating an upper casing of the first embodiment being depressed;
- (9) FIG. 8 is a sectional view taken along line VIII-VIII in FIG. 7;
- (10) FIG. 9 is a sectional view illustrating an insertion seat of the first embodiment being moved toward a host;
- (11) FIG. 10 is a sectional view taken along line X-X in FIG. 9;
- (12) FIG. 11 is a sectional view illustrating a retraction seat of the first embodiment being moved away from the host;
- (13) FIG. 12 is a sectional view taken along line XII-XII in FIG. 11;
- (14) FIG. 13 is a sectional view illustrating an auxiliary insertion seat of the first embodiment being separated from a sensor assembly;
- (15) FIG. 14 is a sectional view taken along line XIV-XIV in FIG. 13;
- (16) FIG. 15 is a sectional view illustrating the lower casing being re-coupled to the upper casing;
- (17) FIG. 16 is a sectional view taken along line XVI-XVI in FIG. 15;
- (18) FIG. 17 is a perspective view illustrating a modification of the insertion device;
- (19) FIG. 18 is a sectional view of the modification shown in FIG. 17, illustrating a stage of being prepared for the implanting;
- (20) FIG. 19 is a sectional view taken along line XIX-XIX in FIG. 18;
- (21) FIG. 20 is another sectional view of the modification, illustrating an instance during the implanting;
- (22) FIG. 21 is a sectional view taken along line XXI-XXI in FIG. 20;
- (23) FIG. 22 is a schematic views illustrating a second embodiment of the insertion device according to the disclosure being assembled;
- (24) FIG. 23 is a schematic view illustrating a lower casing of the second embodiment uncovering a base;
- (25) FIG. 24 is a schematic view illustrating operation of an insertion module;
- (26) FIG. 25 is a schematic view illustrating the lower casing covering the base;
- (27) FIG. 26 is a schematic view illustrating the lower casing tearing a release layer via a tearing element when the lower casing uncovers the base;
- (28) FIG. 27 is a flow chart of an insertion operation of the insertion device according to the disclosure;
- (29) FIG. 28 is a sectional view illustrating a sensor assembly being mounted onto a base;
- (30) FIG. 29 is a partly sectional view illustrating the sensor assembly being connected to an auxiliary insertion seat;
- (31) FIG. 30 is another exploded perspective view of the insertion device;
- (32) FIG. 31 is a sectional view illustrating a third embodiment of the insertion device according to the disclosure;
- (33) FIG. 32 is an enlarged sectional view of the third embodiment; and
- (34) FIG. 33 is a sectional view illustrating a fourth embodiment of the insertion device according to the disclosure.

DETAILED DESCRIPTION

- (35) Before the disclosure is described in greater detail, it should be noted that where considered

appropriate, reference numerals or terminal portions of reference numerals have been repeated among the figures to indicate corresponding or analogous elements, which may optionally have similar characteristics.

(36) Referring to FIGS. **1** to **4**, a first embodiment of the insertion device according to the disclosure is for inserting a sensor into a host. The insertion device may be in the form of a canister (e.g., a desiccant canister), but is not limited to such. The insertion device includes an upper casing **10** and an insertion module **30**.

(37) The upper casing **10** has an accommodating space **14** (see FIG. **4**), and a pair of casing engaging structures **124** (see FIG. **4**).

(38) The insertion module **30** is disposed in the accommodating space **14**, and includes a main body assembly **300**, an insertion seat **33**, a first elastic member **34**, a retraction seat **35**, a second elastic member **37**, a base **50** and a sensor assembly **70**.

(39) With particular reference to FIGS. **3** and **4**, the main body assembly **300** includes an upper portion **302**, and a lower portion **303** that cooperates with the upper portion **302** to define a displacement space **301** therebetween. The lower portion **303** of the main body assembly **300** has a pair of body engaging structures **315** that abut against inner sides of the casing engaging structures **124** of the upper casing **10**, and that are able to respectively engage the casing engaging structures **124** of the upper casing **10** when the upper casing **10** is depressed downwardly.

(40) The insertion seat **33** is removably positioned in the upper portion **302** of the main body assembly **300**, and is able to move in the displacement space **301** between the upper portion **302** and the lower portion **303** along an axial line (L).

(41) The upper casing **10** has a pair of urging portions **123** (see FIG. **4**). The lower portion **303** of the main body assembly **300** has a pair of stopping portions **317** (see FIG. **4**). The insertion seat **33** has a pair of buckle portions **335** (see FIG. **4**) that respectively and separably abut against the stopping portions **317** of the lower portion **303** of the main body assembly **300** so as to position the insertion seat **33** relative to the main body assembly **300**, and that are able to be respectively pushed by the urging portions **123** of the upper casing **10** to be respectively separated from the stopping portions **317** of the lower portion **303** of the main body assembly **300**. The buckle portions **335** of the insertion seat **33** respectively and separably abut the stopping portions **317** of the lower portion **303** of the main body assembly **300** so as to form an insertion limiting structure (A) (see FIG. **4**).

(42) The insertion seat **33** further has at least one limiting groove **330** (see FIG. **4**). The upper portion **302** of the main body assembly **300** has at least one limiting member **323** (see FIG. **4**) that removably engages with the limiting groove **330** of the insertion seat **33**. The insertion seat **33** further has at least one retraction positioning portion **334** that separably abuts against the limiting member **323** of the upper portion **302** of the main body assembly **300**.

(43) The first elastic member **34** has two opposite ends respectively abutting against the insertion seat **33** and the upper portion **302** of the main body assembly **300**. In this embodiment, the first elastic member **34** may be configured as a pre-compressed spring.

(44) The retraction seat **35** is mounted with an insertion needle **36**, and is removably positioned relative to the insertion seat **33**. Specifically, the retraction positioning portion **334** of the insertion seat **33** is removably limited by the limiting member **323** of the upper portion **302** of the main body assembly **300**, so as to form a retraction limiting structure (B) (see FIG. **4**) that positions the retraction seat **35** relative to the insertion seat **33**.

(45) The second elastic member **37** has two opposite ends respectively abutting against the insertion seat **33** and the retraction seat **35**. In this embodiment, the second elastic member **37** may be configured as a pre-compressed spring.

(46) The base **50** is separably positioned relative to the lower portion **303** of the main body assembly **300**, and includes a base seat **51**, and an adhesive pad **52** that is fixedly connected to the base seat **51**. In one embodiment, the base **50** may further include a release layer **55** (see FIGS. **31**

and 33) that is separably adhered to the adhesive pad 52.

(47) The insertion module 30 further includes an auxiliary insertion seat 38 that is separably mounted to the insertion needle 36. The sensor assembly 70 is separably mounted to the auxiliary insertion seat 38. The auxiliary insertion seat 38 has a base portion 381, three angularly spaced-apart wing portions 382 each extending radially outwardly from the base portion 381, and a plurality of coupling portions 383 (see FIG. 3) that protrude from a bottom surface of the base portion 381. Each of the wing portions 382 has a plurality of notches 384, such that the wing portion 382 is flexible in a radial direction of the auxiliary insertion seat 38.

(48) Referring to FIGS. 3 and 4, after first embodiment of the insertion device is assembled, the upper portion 302 and the lower portion 303 of the main body assembly 300 are interconnected, a top portion of the upper casing 10 and a top portion of the upper portion 302 of the main body assembly 300 are spaced apart from each other by a distance (D) (see FIG. 3), the buckle portions 335 of the insertion seat 33 respectively abut against the stopping portions 317 of the lower portion 303 of the main body assembly 300 so as to position the insertion seat 33 at a pre-insertion position, the first elastic member 34 is pre-compressed between the insertion seat 33 and the upper portion 302 of the main body assembly 300 to generate a restoring force, the second elastic member 37 is pre-compressed between the retraction seat 35 and the insertion seat 33 to generate a restoring force, the retraction seat 35 is at a pre-insertion position, the coupling portions 383 of the auxiliary insertion seat 38 is fitted into the coupling portion 711 of the sensing seat 71 so that the sensing seat 71 and the auxiliary insertion seat 38 are interconnected, the auxiliary insertion seat 38 is connected to a needle seat 361 of the insertion needle 36 so that the insertion needle 36 is retained in the lower portion 303 of the main body assembly 300 and is shielded by the base 50, and the base 50 is positioned relative to the lower portion 303 of the main body assembly 300.

(49) The insertion operation performed by the first embodiment (to insert the sensor 72 into the host) is illustrated as follows:

(50) Referring to FIGS. 5 and 6, the adhesive pad 52 is adhered to a skin surface of the host. Referring to FIGS. 7 and 8, when the upper casing 10 is not depressed, the insertion limiting structure (A) that is formed between the buckle portions 335 of the insertion seat 33 and the stopping portions 317 of the lower portion 303 of the main body assembly 300 maintains the insertion seat 33 at the pre-insertion position. When the upper casing 10 is depressed toward the skin surface, the buckle portions 335 of the insertion seat 33 are respectively pushed by the urging portions 123 of the upper casing 10 to be deformed inwardly and to be respectively separated from the stopping portions 317 of the lower portion 303 of the main body assembly 300, so that the insertion limiting structure (A) is collapsed. At the same time, the body engaging structures 315 of the main body assembly 300 respectively engage the casing engaging structure 124 of the upper casing 10, so that the upper casing 10 is positioned relative to the main body assembly 300. After the cover body 12 being pressed by the user, the actuating portions 123 (i.e., the urging portions 123) will be stuck underneath the main body 31, and thus the cover body 12 cannot be moved upward while the user does not feel any vibration nor noise

(51) Referring to FIGS. 9 and 10, after the insertion limiting structure (A) is collapsed, the restoring force of the first elastic member 34 is permitted to be released, and moves the insertion seat 33 to an insertion position to implement automatic-insertion, such that the sensor assembly 70 is moved by the insertion seat 33 to a post-insertion position, that a portion of the sensor 72 is inserted underneath the skin surface, and that the sensing seat 71 is positioned onto the mounting portion 500 of the base 50. After the sensor 72 is inserted underneath the skin surface, the limiting member 323 of the upper portion 302 of the main body assembly 300 is separated from the limiting groove 330 and the retraction positioning portion 334 of the insertion seat 33, so that the retraction positioning portion 334 is permitted to be deformed outwardly to collapse the retraction limiting structure (B). As such, the restoring force of the second elastic member 37 is permitted to be released, and drives the retraction seat 35 to move past the retraction positioning portion 334 of the

insertion seat **33** away from the skin surface, such that the insertion needle **36** is separated from the auxiliary insertion seat **38** and is retracted into the insertion seat **33** to hide a needle **362** thereof and to implement automatic-retraction (see FIGS. **11** and **12**). At this time, the coupling portions **383** of the auxiliary insertion seat **38** is maintained to be fitted into the coupling portion **711** of the sensing seat **71** so that the auxiliary insertion seat **38** is connected to the sensing seat **71** of the sensor assembly **70**.

(52) Afterward, the upper casing **10** and the insertion module **30** are separated from the base **50** and the skin surface of the host (the auxiliary insertion seat **38** is separated from the sensing seat **71** of the sensor assembly **70**), so as to finish the insertion operation of the first embodiment of the insertion device including the automatic-insertion and automatic-retraction operations (see FIGS. **13**, **14** and **28**).

(53) Accordingly, the first embodiment of the insertion device according to the disclosure is operated via a single-action operation. When the upper casing **10** is depressed, the insertion seat **33** is driven by the compressed first elastic member **34** to move toward the skin surface so as to perform the automatic-insertion operation and to separate the limiting structures (i.e., the limiting groove **330** and the limiting member **323**) between the insertion seat **33** and the upper portion **302** of the main body assembly **300**. After the automatic-insertion operation is done, the upper casing **10** is positioned relative to the main body assembly **300** without rebounding, and the limiting structures between the insertion seat **33** and the upper portion **302** are separated from each other, so that the retraction seat **35** is driven by the pre-compressed second elastic member **37** to perform the automatic-retraction operation. In this embodiment, a time needed for performing the automatic-insertion operation and the automatic-retraction operation is no more than 100 milliseconds. In other words, during the insertion operation of the first embodiment, the duration for which the insertion needle **36** is inserted into the host is no more than 100 milliseconds. In other embodiments, the duration may be no more than 50 milliseconds, no more than 8 milliseconds, no more than 6 milliseconds, no more than 4 milliseconds or no more than 2 milliseconds.

(54) Referring back to FIGS. **1** to **4**, a modification of the first embodiment of the insertion device according to the disclosure is for inserting a sensor into a host. The modification may also be in the form of a canister (e.g., a desiccant canister), but is not limited to such.

(55) The modification of the insertion device includes an upper casing **10**, a lower casing **20**, an insertion module **30**, a base **50** and a desiccant **60**.

(56) The upper casing **10** includes an outer shell **11**, and a cover body **12** that is disposed in the outer shell **11**. The cover body **12** has at least one casing engaging structure **124** at a lower portion thereof. The outer casing **11** has an annular protruding portion **115** and an upper positioning portion **116**.

(57) The lower casing **20** is separably and airtightly connected to the upper casing **10** so as to form an airtight space **200** (see FIG. **4**) therebetween. The desiccant **60** is disposed in the airtight space **200**.

(58) In this modification, the upper casing **10** and the lower casing **20** are interconnected in a hard-interference manner, but are not limited to such. The lower casing **20** has a top edge **233**. The top edge **233** of the lower casing **20** is provided with a lower positioning portion **227** that corresponds in position and shape to the upper positioning portion **116** of the upper casing **10**. The lower casing **20** further has at least one abutment portion **228** at an inner surrounding surface thereof that abuts against an outer side of the casing engaging structure **124** of the upper casing **10**, and an annular groove **224** at the inner surrounding surface thereof that permits the annular protruding portion **115** of the upper casing **10** to engage therewith so as to interconnect the upper casing **10** and the lower casing **20**.

(59) In this modification, the insertion device may further include a sealing ring **13** (see FIGS. **3** and **4**) that is disposed between the annular protruding portion **115** of the upper casing **10** and the annular groove **224** of the lower casing **20** for enhancing air-tightness of the airtight space **200**.

defined between the upper casing **10** and the lower casing **20**.

(60) The insertion module **30** is disposed in the upper casing **10**, and is able to be moved by the upper casing **10**. The insertion module **30** includes a main body assembly **300**. The main body assembly **300** includes a main body **31**, and a main cover **32** that is connected to the main body **31** and that cooperates with the main body **31** to define a displacement space **301** therebetween. The main body assembly **300** further has at least one body engaging structure **315** that abuts against an inner side of the casing engaging structure **124** of the upper casing **10**. The body engaging structure **315** engages with the casing engaging structure **124** of the upper casing **10** when the upper casing **10** is depressed.

(61) In this modification, the body engaging structure **315** of the main body assembly **300** is disposed on the main body **31** of the main body assembly **300**, and the body engaging structure **315** abuts against the inner side of the casing engaging structure **124** of the cover body **12** of the upper casing **10**. When the lower casing **20** engages the upper casing **10**, the abutment portion **228** of the lower casing **20** limits movement of the casing engaging structure **124** of the cover body **12** of the upper casing **10**, such that the upper casing **10** cannot move downwardly so as to prevent unintentional insertion operation of the insertion device due to unintentionally applying force to the insertion device.

(62) The insertion module **30** further includes an insertion seat **33** that is disposed in the displacement space **301**, a first elastic member **34** that has two opposite ends respectively abutting against the insertion seat **33** and the main cover **32** for moving the insertion seat **33** in an insertion direction (F), a retraction seat **35** that is disposed in the insertion seat **33**, an insertion needle **36** that is connected to the retraction seat **35**, a second elastic member **37** that has two opposite ends respectively abutting against the insertion seat **33** and the retraction seat **35** for moving the retraction seat **35** in a retraction direction (R) opposite to the insertion direction (F), an auxiliary insertion seat **38** that is separably connected to the insertion needle **36**, and a sensor assembly **70** that is separably connected to the auxiliary insertion seat **38**. The sensor assembly **70** includes a sensing seat **71**, and a sensor **72** that is mounted to the sensing seat **71** and that is separably coupled to the insertion needle **36**.

(63) In this modification, the first elastic member **34** and the second elastic **37** are configured as pre-compressed springs.

(64) The cover body **12** of the upper casing **10** has a pair of urging portions **123** at an inner surrounding surface thereof. The insertion seat **33** has a pair of buckle portions **335** (see FIG. 4) that respectively and separably abut against two stopping portions **317** of the main body **31**, and that are able to be respectively pushed by the urging portions **123** of the cover body **12** to be respectively separated from the stopping portions **317** of the main body **31**. The buckle portions **335** of the insertion seat **33** respectively and separably abut the stopping portions **317** of the main body **31** so as to form an insertion limiting structure (A) between the insertion seat **33** and the main body **31**.

(65) The insertion seat **33** further has at least one limiting groove **330**. The main cover **32** has at least one limiting member **323** that removably engages with the limiting groove **330** of the insertion seat **33**. The insertion seat **33** further has at least one retraction positioning portion **334** that separably abuts against the limiting member **323** of the main cover **32** and that limits movements of the retraction seat **35**. The retraction positioning portion **334** of the insertion seat **33** is limited by the limiting member **323** of the main cover **32**, so as to form a retraction limiting structure (B) among the retraction seat **35**, the insertion seat **33** and the main cover **32** of the main body assembly **300**.

(66) In this embodiment, the sensor assembly **70** is to be separably mounted to the base **50**, and includes a sensing seat **71**, and a sensor **72** that is mounted to the sensing seat **71** and that is separably coupled to the insertion needle **36**. The sensing seat **71** has a coupling portion **711** (see FIG. 3).

(67) The base **50** is separably positioned relative to the main body **32** of the main body assembly **300**, and permits the sensor assembly **70** to be mounted thereon after the sensor assembly **70** is separated from the auxiliary insertion seat **38**. The base **50** includes a base seat **51**, an adhesive pad **52** that is fixedly connected to the base seat **51**, and two base hook structures **54** (see FIG. 3). In one embodiment, the base **50** may further include a release layer **55** (see FIGS. 31 and 33) that is separably adhered to the adhesive pad **52**. The base **50** further has a mounting portion **500** (see FIG. 4) for the sensor assembly **70** to be separably mounted thereto. In this modification, the mounting portion **500** of the base **50** includes a recess **501** that permits the sensor assembly **70** to be press-fitted therein, or may include an adhesive layer **502** (e.g., double-sided tape) that is disposed in the recess **501** (see FIG. 31), or at least one resilient hook **503** (see FIG. 32) that is for engaging with the sensor assembly **70**.

(68) In this modification, the insertion device further includes a pair of fixing members **40**. The fixing members **40** are mounted to a slide groove **310** (see FIG. 2) of the main body **31** of the main body assembly **300**. Each of the fixing members **40** has a push portion **41** that corresponds in position to a respective one of two stop portions **226** (see FIG. 3) of the lower casing **20**, a support portion **42** that is opposite to the push portion **41**, a fixing hook structure **43** that is located between the push portion **41** and the support portion **42**, and a cam portion **44** that is located between the push portion **41** and the support portion **42** and that has an inclined cam surface **441**.

(69) The push portion **41** of each of the fixing members **40** is limited by the respective one of the stop portions **226** of the lower casing **20**, so that the fixing members **40** are positioned relative to the main body **31**. The sensing seat **71** of the sensor assembly **70** is supported by the support portions **42** of the fixing members **40** so as to be positioned relative to the main body **31**. The fixing hook structures **43** of the fixing members **40** respectively engage with the base hook structures **54** of the base **50**.

(70) With reference to FIGS. 3 and 4, before the lower casing **20** is separated from the upper casing **10**, the fixing members **40** interfere with the lower casing **20** and the cover body **12** of the upper casing **10**, so that the upper casing **10** is prevented from moving downwardly by the fixing members **40** to further prevent unintentional insertion operation of the insertion device.

(71) Since the sensor assembly **70** has a lower end supported by the fixing members **40**, and since the fixing members **40** engage with the base **50**, the base **50** is limited within the insertion device. When the lower casing **20** is separated from the upper casing **10** (or uncovers the base **50** while being connected to the upper casing **10**, in a modification), the upper casing **10** is permitted to be depressed, such that the cover body **12** moves downwardly to push and move the fixing member **40** laterally and outwardly so as to collapse the limiting structure among the fixing members **40**, the sensor assembly **70** and the base **50**. As such, the first elastic member **34** of the insertion module **30** is permitted to release the restoring force thereof to insert the insertion needle **36** and the sensor **72** of the sensor assembly **70** underneath the skin surface of the host, and the second elastic member **37** of the insertion module **30** is permitted to release the restoring force thereof to retract the insertion needle **36** to finish automatic-insertion and automatic-retraction operations.

(72) Referring to FIGS. 3, 4, and 22 to 26 (FIGS. 22 to 26 are schematic views), a second embodiment, or a modification of the first embodiment of the insertion device according to the disclosure, is for inserting a sensor into a host. The outer shell **11** and the lower casing **20** (or the upper casing **20** and the lower casing **20**) are airtightly coupled to each other, so that an inner space of the insertion device is airtight. The desiccant **60** may be disposed at a suitable position anywhere in the inner space of the insertion device. In another modification, any one of the components of the insertion module **30** may be made of materials including desiccant through an injection molding technology (as shown in FIGS. 22 and 23). In still another modification, an inner surface of the outer shell **11** may be provided with a desiccant layer **62** (see FIGS. 25 and 26), or the sensor assembly **70** may have a desiccant **60** (not shown) so that the inner space of the insertion device is kept dry and that the sensor assembly **70** is prevented from being moistened.

(73) Referring further to FIGS. 25 and 26, the base 50 further includes a tearing element 23 that is fixedly mounted to a base plate 21 of the lower casing 20. The tearing element 23 is connected to the release layer 55 (see FIGS. 31 and 33) that is separably adhered to the adhesive pad 52. When the lower casing 20 is separated from the upper casing 10 (or uncovers the base 50 while being connected to the upper casing 10), the tearing element 23 simultaneously separates the release layer 55 from the adhesive pad 52. Since the release layer 55 is not separated from the adhesive pad 52 until the insertion device is to perform the insertion operation, the adhesive pad 52 is prevented from being stained before the insertion operation. In this embodiment, the lower casing 20 is in the form of a cap. In other embodiments, the lower casing 20 may be in the form of an aluminum foil (not shown).

(74) Referring to FIGS. 3 and 4, the outer shell 11 is sleeved onto the insertion module 30 after the components of the insertion module 30 are assembled. At this time, the sensor assembly 70 has been mounted to the insertion module 30 via the auxiliary insertion seat 38 (see FIG. 29), the lower casing 20 is subsequently coupled to the outer shell 11. The sensor assembly 70 does not need to be manually adjusted during the insertion operation of the insertion device. It means that a user does not require additional manual operation of grasping the sensor assembly 70 onto the base 50 by the implanting module 30 (i.e., the insertion module 30). In this embodiment, all the components of the insertion device have been assembled in the factory, but can be disassembled and reassembled by medical personnel or a user (see FIG. 30). When the insertion device is used to perform the insertion operation, the user can easily manipulate the insertion device (e.g., by simply depressing the upper casing 10) to mount the base 50 onto the skin surface of the host by the adhesive pad 52 of the base 50 followed by coupling the sensor assembly 70 onto the base 50.

(75) Referring to FIGS. 15 and 16, the separated lower casing 20 can be re-coupled to the upper casing 10. The lower casing 20 can seal a bottom portion of the outer shell 11 by the engagement between the annular protruding portion 115 of the upper casing 10 and the annular groove 224 of the lower casing 20. When the lower casing 20 is re-coupled to the outer shell 11 of the upper casing 10, the engagement between the lower positioning portion 227 of the lower casing 20 and the upper positioning portion 116 of the upper casing 10 ensures that the lower casing 20 and the upper casing 10 are properly positioned to each other. In addition, the inner space of the insertion device can serve as a waste storing space. A used base 50 can be removed from the host and be placed in the inner space of the insertion device, such that the used insertion device can be disposed in compliance with standard of discard of medical waste. The whole insertion operation of the insertion device according to the disclosure is shown in FIG. 27.

(76) FIG. 28 shows that after the sensor 72 is implanted under the skin surface P of the living body, the sensor assembly 70 and the base 50 that are simultaneously disposed on the skin surface P of the living body, a transmitter 90 is installed on the base 50 in the form of an upper cover to process the physiological signals measured by the sensor 72 and allows the signals to be transmitted to the outside. In order to reduce the number of components to be implanted during the implantation process and also reduce the loading of the container, the inserter device of the present invention does not include a transmitter. The separately arranged the transmitter 90 and the sensor assembly 70 are to ensure the electronic parts are not damaged by the sterilization process, so the production yield of the device can be improved.

(77) Referring to FIGS. 31 and 32, a third embodiment of the insertion device (similar to the automatic needle insertion and automatic needle extraction of the first embodiment, but in this embodiment does not include the outer shell 11) can optionally have a bottom cover 20, or uses a sealed package to seal the insertion device. The insertion device includes a cover body 12 and an insertion module 30.

(78) The cover body 12 has an accommodating space 14, and a pair of casing engaging structures 124.

(79) The insertion module 30 is disposed in the accommodating space 14, and includes a main

body **31**, an insertion seat **33**, a first elastic member **34**, a retraction seat **35**, a second elastic member **37**, a base **50** and a sensor assembly **70**.

(80) The main body **31** is associated with the cover body **12**, and cooperates with the cover body **12** to define a displacement space **301** therebetween. The main body **31** has a pair of body engaging structures **315** that are able to respectively engage the casing engaging structure **124** of the cover body **12**.

(81) The insertion seat **33** is removably positioned in the cover body **12**, and is able to move in the displacement space **301** between the main body **31** and the cover body **12** along an axial line (L).

(82) The cover body **12** has a pair of urging portions **123** on an inner surrounding surface thereof. The main body **31** has a pair of stopping portions **317**. The insertion seat **33** has a pair of buckle portions **335** that respectively and separably abut against the stopping portions **317** of the main body **31** so as to position the insertion seat **33** relative to the main body **31**, and that are able to be respectively pushed by the urging portions **123** of the cover body **12** to be respectively separated from the stopping portions **317** of the main body **31**. The buckle portions **335** of the insertion seat respectively and separably abut the stopping portions **317** of the main body **31** so as to form an insertion limiting structure (A2) between the insertion seat **33** and the main body **31**.

(83) The insertion seat **33** further has at least one limiting groove **330**. The cover body **12** further has at least one limiting member **100** that removably engages with the limiting groove **330** of the insertion seat **33**. The insertion seat **33** further has at least one retraction positioning portion **334** that separably abuts against the limiting member **100** of the cover body **12**.

(84) The first elastic member **34** has two opposite ends respectively abutting against the insertion seat **33** and the cover body **12**. In this embodiment, the first elastic member **34** may be configured as a pre-compressed spring.

(85) The retraction seat **35** is mounted with an insertion needle **36**, and is removably positioned relative to the insertion seat **33**. The retraction positioning portion **334** of the insertion seat **33** is removably limited by the limiting member **100** of the cover body **12**, so as to form a retraction limiting structure (B2) that positions the retraction seat **35** relative to the insertion seat **33**.

(86) The second elastic member **37** has two opposite ends respectively abutting against the insertion seat **33** and the retraction seat **35**. In this embodiment, the second elastic member **37** may be configured as a pre-compressed spring.

(87) The base **50** is separably positioned relative to the main body **31**, and includes a base seat **51**, and an adhesive pad **52** that is fixedly connected to the base seat **51**.

(88) The sensor assembly **70** is to be separably mounted to the base **50**, and includes a sensing seat **71**, and a sensor **72** that is mounted to the sensing seat **71** and that is separably coupled to the insertion needle **36**. The base **50** further has a mounting portion **500** for the sensor assembly **70** to be separably mounted thereto, and the mounting methods are similar as those described in the previous embodiment. The insertion module **30** further includes an auxiliary insertion seat **38** that is separably mounted to the insertion needle **36**. The sensor assembly **70** is separably mounted to the auxiliary insertion seat **38**. The auxiliary insertion seat **38** slidably engages with the main body assembly **300** to guide the insertion needle **36** to steadily insert the sensor **72** into the host without runout or inclination so as to alleviate pain of the host.

(89) Referring to FIG. **33**, a fourth embodiment of the insertion device according to the disclosure (similar to the third embodiment but the insertion module includes independent main cover) includes a cover body **12** and an insertion module **30**.

(90) The cover body **12** has an accommodating space **14**, and a pair of casing engaging structures **124**.

(91) The insertion module **30** is disposed in the accommodating space **14**, and includes a main body **31**, a main cover **32**, an insertion seat **33**, a first elastic member **34**, a retraction seat **35**, a second elastic member **37**, a base **50** and a sensor assembly **70**.

(92) The main body **31** is connected to the main cover **32**, and cooperates with the main cover **32** to

define a displacement space **301** therebetween. The main body **31** has a pair of body engaging structures **315** that are able to respectively engage the casing engaging structure **124** of the cover body **12**. A top portion of the cover body **12** and a top portion of the main cover **32** is spaced apart from each other by a distance (D).

(93) The insertion seat **33** is removably positioned in the main cover **32**, and is able to move in the displacement space **301** between the main body **31** and the main cover **32** along an axial line (L).

(94) The cover body **12** has a pair of urging portions **123** on an inner surrounding surface thereof. The main body **31** has a pair of stopping portions **317**. The insertion seat **33** has a pair of buckle portions **335** that respectively and separably abut against the stopping portions **317** of the main body **31** so as to position the insertion seat **33** relative to the main body **31**, and that are able to be respectively pushed by the urging portions **123** of the cover body **12** to be respectively separated from the stopping portions **317** of the main body **31**. The buckle portions **335** of the insertion seat **33** respectively and separably abut the stopping portions **317** of the main body **31** so as to form an insertion limiting structure (A1) between the insertion seat **33** and the main body **31**.

(95) The insertion seat **33** further has at least one limiting groove **330**. The main cover **32** further has at least one limiting member **320** that removably engages with the limiting groove **330** of the insertion seat **33**. The insertion seat **33** further has at least one retraction positioning portion **334** that separably abuts against the limiting member **320** of the main cover **32**.

(96) The first elastic member **34** has two opposite ends respectively abutting against the insertion seat **33** and the main cover **32**. In this embodiment, the first elastic member **34** may be configured as a pre-compressed spring.

(97) The retraction seat **35** is mounted with an insertion needle **36**, and is removably positioned relative to the insertion seat **33**. The retraction positioning portion **334** of the insertion seat **33** is removably limited by the limiting member **320** of the main cover **32**, so as to form a retraction limiting structure (B1) that positions the retraction seat **35** relative to the insertion seat **33**.

(98) The second elastic member **37** has two opposite ends respectively abutting against the insertion seat **33** and the retraction seat **35**. In this embodiment, the second elastic member **37** may be configured as a pre-compressed spring.

(99) The base **50** is separably positioned relative to the main body **31**, and includes a base seat **51**, and an adhesive pad **52** that is fixedly connected to the base seat **51**.

(100) The sensor assembly **70** is to be separably mounted to the base **50**, and includes a sensing seat **71**, and a sensor **72** that is mounted to the sensing seat **71** and that is separably coupled to the insertion needle **36**. The base **50** further has a mounting portion **500** for the sensor assembly **70** to be separably mounted thereto, and the mounting methods are similar as those described in the previous embodiment.

(101) The insertion module **30** further includes an auxiliary insertion seat **38** that is separably mounted to the insertion needle **36**. The sensor assembly **70** is separably mounted to the auxiliary insertion seat **38**. The auxiliary insertion seat **38** slidably engages with the main body assembly **300** to guide the insertion needle **36** to steadily insert the sensor **72** into the host without runout or inclination so as to alleviate pain of the host.

(102) The insertion device according to the disclosure (the first to fourth embodiments and the modifications thereof) is operated in such a manner that the upper casing **10** or the cover body **12** is depressed so as to collapse the insertion limiting structure (A, A1, A2), and the first elastic member **34** therefore releases the restoring force to move the insertion seat **33** downwardly to perform the automatic-insertion operation. In other words, in this disclosure, the insertion needle **36** is driven to be inserted into the host by the first elastic member **34** rather than by manual depression of a user. Therefore, the insertion operation and the retraction operation can be smoothly performed by any user even if the user is not an adept operator.

(103) The insertion operation performed by the third and fourth embodiments, according to the disclosure are operated via a single-action operation. The insertion operation of the sensor **72** into

the host is illustrated as follows:

(104) Referring to FIGS. **31** to **33**, when the adhesive pad **52** is adhered to a skin surface of the host and the cover body **12** is not depressed, the insertion limiting structure (**A1**, **A2**) that is formed between the buckle portions **335** of the insertion seat **33** and the stopping portions **317** of the main body **31** maintains the insertion seat **33** at a pre-insertion position. When the cover body **12** is depressed toward the skin surface, the buckle portions **335** of the insertion seat **33** are respectively pushed by the urging portions **123** of the cover body **12** to be deformed inwardly and to be respectively separated from the stopping portions **317** of the main body **31**, so that the insertion limiting structure (**A1**, **A2**) is collapsed. At the same time, the body engaging structures **315** of the main body **31** respectively engage the casing engaging structure **124** of the cover body **12**, so that the cover body **12** is positioned relative to the main body **31**.

(105) After the insertion limiting structure (**A1**, **A2**) is collapsed, the restoring force of the first elastic member **34** is permitted to be released, and moves the insertion seat **33** to an insertion position to implement automatic-insertion, such that the sensor assembly **70** is moved by the insertion seat **33** to a post-insertion position, that the sensor **72** is inserted underneath the skin surface, and that the sensing seat **71** is positioned onto the mounting portion **500** of the base **50**. After the sensor **72** is inserted underneath the skin surface, the limiting member **320** of the main cover **32** or the limiting member **100** of the cover body **12** is separated from the limiting groove **330** and the retraction positioning portion **334** of the insertion seat **33**, so that the retraction positioning portion **334** is permitted to be deformed outwardly to collapse the retraction limiting structure (**B1**, **B2**). As such, the restoring force of the second elastic member **37** is permitted to be released, and drives the retraction seat **35** to move past the retraction positioning portion **334** of the insertion seat **33** away from the skin surface, such that the insertion needle **36** is separated from the auxiliary insertion seat **38** and is retracted into the insertion seat **33** to hide a needle **362** thereof and to implement automatic-retraction. At this time, the coupling portions **383** of the auxiliary insertion seat **38** is maintained to be fitted into the coupling portion **711** of the sensing seat **71** so that the auxiliary insertion seat **38** is connected to the sensing seat **71** of the sensor assembly **70**.

(106) Afterward, the cover body **12** and the insertion module **30** are separated from the base **50** and the skin surface of the host, so as to finish the insertion operation of the third and fourth embodiments of the insertion device including the automatic-insertion and automatic-retraction operations.

(107) Accordingly, the third and fourth embodiments of the insertion device may further include a lower casing (not shown) that is air-tightly and separably coupled to the cover body **12** so as to form an airtight space within the insertion device. A desiccant may be provided in the inner space of the insertion device so as to prevent the sensor **72** of the sensor assembly **70** from being moistened.

(108) In the third or fourth embodiments, a time needed for performing the automatic-insertion operation and the automatic-retraction operation (i.e., the duration for which the insertion needle **36** is inserted into the host) is no more than 100 milliseconds, no more than 50 milliseconds, no more than 8 milliseconds, no more than 6 milliseconds, no more than 4 milliseconds or no more than 2 milliseconds;

(109) In the fourth embodiment, by virtue of the main cover that cooperates with the insertion seat **33** to compress the first elastic member **34** therebetween, the restoring force of the first elastic member **34** would not act on the cover body **12** when a user depresses the cover body **12**, rendering the operation of the insertion device smooth. The insertion module further includes an insertion seat **33** and a retraction seat **35**. The insertion limiting structure (**A1**, **A2**) is formed between the insertion seat **33** and the main body **31**. The retraction limiting structure (**B1**, **B2**) is formed among the insertion seat **33**, the retraction seat **35** and the main cover **31** or among the insertion seat **33**, the retraction seat **35** and the cover body **12**.

(110) The retraction seat **35** includes an insertion needle **36**. The insertion seat **33** includes a

pushing portion (with reference to the reference numeral **337** in FIG. 3). After the automatic-insertion operation performed by the insertion module **30**, the insertion needle **36** is inserted through the base **50** to insert a sensor **72** underneath a skin surface of a host, and the pushing portion of the insertion seat **33** tightly abuts against the sensor assembly **70** (via an auxiliary insertion seat **38**) so as to prevent the sensor assembly **70** from being separated from a mounting portion **500** of the base **50**.

(111) The advantages of the insertion device according to the disclosure are as follows:

(112) 1. The insertion device employs the pre-compressed first and second elastic members **34**, **37** to perform the automatic-insertion and automatic-retraction operations, so that insertion device is operated via a single-action operation (i.e., by depressing the upper casing **10**, outer shell **11** or the cover body **12**) to finish the automatic-insertion and automatic-retraction operations. The insertion needle **36** is not driven to be inserted into the host by manual depression of a user. Therefore, the insertion operation and the retraction operation can be smoothly performed by any user even if the user is not an adept operator.

(113) 2. The lower casing **20** cooperates with the upper casing **10** or the cover body **12** to form the airtight space that is provided with the desiccant **60** therein, so that the sensor **72** of the sensor assembly **70** is prevented from being moistened to ensure the accuracy of the sensor assembly **70**. In addition, a desiccation indicator **61** (see FIG. 4) may be provided on inner or outer side of the outer shell **11** of the upper casing **10**. Preferably, the desiccation indicator **61** may be an aridity indication material. The outer shell **11** may have a transparent or translucent portion through which the desiccation indicator **61** is visible, so a user can perceive whether or not the sensor assembly **70** is moistened. The manner in which the desiccant **60** is disposed is not limited to the disclosed embodiments, and the upper casing **10** and the lower casing **20** are not limited to be interconnected in a hard-interference manner.

(114) 3. Before the lower casing **20** is separated (or opened) from the upper casing **10** or from the cover body **12**, the abutment portion **228** of the lower casing **20** limits movement of the casing engaging structure **124** of the cover body **12** or the upper casing **10**, such that the upper casing **10** or the cover body **12** cannot move downwardly so as to prevent unintentional insertion operation of the insertion device due to unintentionally applying force to the insertion device.

(115) 4. After the automatic-retraction operation, the insertion needle **36** is separated from the auxiliary insertion seat **38** and is retracted into the insertion seat **33** to hide a needle **362**.

(116) 5. By virtue of the main cover **32** that cooperates with the insertion seat **33** to compress the first elastic member **34** therebetween, the restoring force of the first elastic member **34** would not act on the upper casing **10** or the cover body **12** when a user depresses the upper casing **10** or the cover body **12**, rendering the operation of the insertion device smooth.

(117) 6. The auxiliary insertion seat **38** slidably engages with the main body assembly **300** to guide the insert direction of the insertion needle **36** to steadily insert the sensor **72** into the host without runout or inclination so as to alleviate pain of the host.

(118) 7. The base **50** has a mounting portion **500** for the sensor assembly **70** to be separably mounted thereto. The mounting portion **500** may include a recess **501** that permits the sensor assembly **70** to be press-fitted therein, and/or an adhesive layer **502** (e.g., double-sided tape) that is disposed in the recess **501** for the sensor assembly **70** to be adhered thereto, or at least one resilient hook **503** that is for engaging with the sensor assembly **70**.

(119) 8. After the upper casing **10** (or housing **11**) or the cover body **12** is depressed, a time needed for performing the automatic-insertion operation and the automatic-retraction operation is no more than 100 milliseconds, no more than 50 milliseconds, no more than 8 milliseconds, no more than 6 milliseconds, no more than 4 milliseconds or no more than 2 milliseconds.

(120) 9. In assembling the insertion device, the first elastic member **34**, the retraction seat **35**, the second elastic member **37** and the insertion seat **33** are firstly disposed between the main cover **32** and the main body **31**, and the auxiliary insertion seat **38** and the sensor assembly **70** are mounted

to the insertion needle **36** before the insertion needle **36** is mounted to the retraction seat **35**. Whereby the sensor assembly **70** and the insertion module **30** forming a clutch design, which can not only greatly improve the assembly yield, but also effectively reduce the cost of the sensor assembly.

(121) 10. The sensor assembly **70** has been mounted to the insertion module **30** via the auxiliary insertion seat **38** (see FIG. **29**), so the sensor assembly **70** does not require additional manual operation of grasping the sensor assembly **70** onto the base **50** by the implanting module **30**.

(122) 11. An inner side of a bottom portion of the upper casing **10** may be provided with a cushion wall **80** (see FIGS. **17** to **21**) that is substantially ring-shaped. Before the insertion device is operated, the base **50** is higher than a bottom edge of the cushion wall **80**. When the bottom edge of the cushion wall **80** abuts against the skin surface of the host, the base **50** is spaced apart from the skin surface, so a user can move the insertion device along the skin surface to a desired position, and depress the upper casing **10**. The cushion wall **80** retracts into the upper casing **10** upon the depression, such that the base **50** can be mounted onto the skin surface. By virtue of the cushion wall **80**, the insertion device can be moved along the skin surface to a desired position with the cushion wall **80** abutting against the skin surface, rendering the insertion operation stable.

(123) 12. When the lower casing **20** is re-coupled to the outer shell **11** of the upper casing **10**, the engagement between the lower positioning portion **227** of the lower casing **20** and the upper positioning portion **116** of the upper casing **10** ensures that the lower casing **20** and the upper casing **10** are properly positioned to each other.

(124) In addition to the embodiments described above, this disclosure further discloses a plurality of embodiments as defined by the claims, with each embodiment comprising the claim elements of the respective claim and the claim elements of any claim upon which the respective claim depends.

(125) In the description above, for the purposes of explanation, numerous specific details have been set forth in order to provide a thorough understanding of the embodiments. It will be apparent, however, to one skilled in the art, that one or more other embodiments may be practiced without some of these specific details. It should also be appreciated that reference throughout this specification to “one embodiment,” “an embodiment,” an embodiment with an indication of an ordinal number and so forth means that a particular feature, structure, or characteristic may be included in the practice of the disclosure. It should be further appreciated that in the description, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of various inventive aspects, and that one or more features or specific details from one embodiment may be practiced together with one or more features or specific details from another embodiment, where appropriate, in the practice of the disclosure.

(126) While the disclosure has been described in connection with what are considered the exemplary embodiments, it is understood that this disclosure is not limited to the disclosed embodiments but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

Claims

1. An insertion device comprising: a cover body having an accommodating space; and an insertion module disposed in said accommodating space, and including a main body, an insertion seat, a first elastic member, a retraction seat, a second elastic member, a base and a sensor assembly, said main body being coupled to said cover body, and cooperating with said cover body to define a displacement space therebetween, said insertion seat being removably positioned in said cover body, and being able to move in said displacement space between said main body and said cover body, said first elastic member having two opposite ends that respectively abut against said insertion seat and said cover body, said retraction seat being removably positioned relative to said

insertion seat, said second elastic member having two opposite ends that respectively abut against said insertion seat and said retraction seat, said base being separably positioned relative to said main body, said sensor assembly being separated from said base before an automatic-insertion operation, and being driven by said insertion seat to be mounted to said base after the automatic-insertion operation, said base having a mounting portion for said sensor assembly to be separably mounted thereto, said mounting portion of said base including a recess that permits said sensor assembly to be press-fitted thereinto; wherein, when said cover body is depressed, the insertion seat is driven by a restoring force of said first elastic member to perform the automatic-insertion operation and to collapse a first limiting structure between said insertion seat and said main body; and wherein, after the automatic-insertion operation is completed, said cover body is positioned relative to said main body without rebounding, and a second limiting structure between said insertion seat and said retraction seat collapses upon the collapse of said first limiting structure between said insertion seat and said main body, so that said retraction seat is driven by a restoring force of said second elastic member to perform an automatic-retraction operation, wherein said insertion seat has at least one limiting groove, said cover body has at least one limiting member that removably engages with said at least one limiting groove of said insertion seat, and said insertion seat further has at least one positioning portion that positions said retraction seat relative to said insertion seat, and wherein, when the at least one limiting member of the cover body is disposed in said at least one limiting groove of the insertion seat, a first side of said at least one limiting member directly abuts against said at least one positioning portion of the insertion seat and a second side of said at least one limiting member directly abuts against an opposing portion of the insertion seat such that removal of said at least one limiting member from the at least one limiting groove directly causes the at least one positioning portion to undergo outward deformation into the at least one limiting groove, wherein the outward deformation of the at least one positioning portion directly releases the restoring force of said second elastic member to perform the automatic-retraction operation.

2. The insertion device as claimed in claim 1, wherein said retraction seat is mounted with an insertion needle.
3. The insertion device as claimed in claim 2, wherein said insertion module further includes an auxiliary insertion seat that is separably mounted to said insertion needle, said sensor assembly being separably mounted to said auxiliary insertion seat.
4. The insertion device as claimed in claim 1, wherein said at least one limiting groove cooperates with said at least one limiting member and said at least one positioning portion to form a retraction limiting structure among said insertion seat, said retraction seat and said cover body that serves as said second limiting structure.
5. The insertion device as claimed in claim 1, wherein said cover body has at least one urging portion on an inner surrounding surface thereof, said main body having at least one stopping portion, said insertion seat having at least one buckle portion that separably abuts against said at least one stopping portions of said main body so as to position said insertion seat relative to said main body, said buckle portion being able to be pushed by said at least one urging portion of said cover body to be separated from said at least one stopping portion of said main body, said at least one buckle portion of said insertion seat cooperating with said at least one stopping portion of said main body to form an insertion limiting structure between said insertion seat and said main body that serves as said first limiting structure.
6. The insertion device as claimed in claim 1, wherein each of said first elastic member and said second elastic member is configured as a pre-compressed spring.
7. The insertion device as claimed in claim 1, further comprising a lower casing that is air-tightly and separably coupled to said cover body to form an airtight space within said insertion device, said airtight space being provided with a desiccant therein.
8. The insertion device as claimed in claim 1, wherein said cover body has at least one casing

engaging structure, said main body having at least one body engaging structure that is able to engage said at least one casing engaging structure of said cover body, said body engaging structure of said main body being operated to engage said casing engaging structure of said cover body when said cover body is depressed toward a skin surface of a host to perform the automatic-insertion operation, so that said cover body is positioned relative to said main body without rebounding after the automatic-insertion operation is completed.

9. The insertion device as claimed in claim 1, wherein said recess of said mounting portion of said base is provided with an adhesive layer therein for said sensor assembly to be adhered thereto, or at least one resilient hook that is for engaging with said sensor assembly.

10. The insertion device as claimed in claim 1, wherein a time needed for performing the automatic-insertion operation and the automatic-retraction operation is no more than 100 milliseconds, no more than 8 milliseconds, no more than 4 milliseconds or no more than 2 milliseconds.

11. An insertion method of a sensor, comprising: an insertion device including a displacement space that is cooperatively defined by a cover body and a main body; an insertion module disposed in the displacement space, the insertion module being movable in the displacement space by compressed first and second elastic members therein so as to perform an automatic-insertion operation and an automatic-retraction operation, wherein the insertion module includes an insertion seat and a retraction seat; an insertion limiting structure disposed between the insertion module and the main body for removably positioning the insertion module at a pre-insertion position; and a retraction limiting structure disposed between the insertion module and the cover body, a portion of the insertion module being able to move from a post-insertion position to the pre-insertion position when the retraction limiting structure is collapsed; wherein, when the insertion device is kept being depressed, the insertion limiting structure is collapsed, and the insertion module is biased by a restoring force of the first elastic member to move downwardly to perform the automatic-insertion operation, the retraction limiting structure being collapsed immediately after the insertion module finishing the automatic-insertion operation, such that the insertion module is biased by a restoring force of the second elastic member to move upwardly to perform the automatic-retraction operation, the insertion module positioning a sensor assembly thereof onto a base during the automatic-insertion operation, a time needed for performing the automatic-insertion operation and the automatic-retraction operation being no more than 100 milliseconds; wherein the sensor assembly is separated from the base before the automatic-insertion operation, and is driven to be mounted to the base after the automatic-insertion operation; and wherein the base has a mounting portion for the sensor assembly to be separably mounted thereto, the mounting portion of the base including a recess that permits the sensor assembly to be press-fitted thereinto, wherein said insertion seat has at least one limiting groove, said cover body has at least one limiting member that removably engages with said at least one limiting groove of said insertion seat, and said insertion seat further has at least one positioning portion that positions said retraction seat relative to said insertion seat, and wherein, when the at least one limiting member of the cover body is disposed in said at least one limiting groove of the insertion seat, a first side of said at least one limiting member directly abuts against said at least one positioning portion of the insertion seat and a second side of said at least one limiting member directly abuts against an opposing portion of the insertion seat such that removal of said at least one limiting member from the at least one limiting groove directly causes the at least one positioning portion to undergo outward deformation into the at least one limiting groove, wherein the outward deformation of the at least one positioning portion directly releases the restoring force of said second elastic member to perform the automatic-retraction operation.

12. The insertion method as claimed in claim 11, wherein the insertion module includes an insertion seat and a retraction seat, the insertion limiting structure being formed between the insertion seat and the main body, the retraction limiting structure being formed among the insertion

seat, the retraction seat and the cover body, the retraction seat being mounted with an insertion needle, the insertion seat having a pushing portion.

13. The insertion method as claimed in claim 11, wherein the recess of the mounting portion of the base is provided with an adhesive layer therein for the sensor assembly to be adhered thereto, or at least one resilient hook that is for engaging with the sensor assembly.

14. The insertion method as claimed in claim 11, wherein, after the automatic-insertion operation performed by the insertion module, an insertion needle is inserted through the base to insert a sensor underneath a skin surface of a host, and a pushing portion of an insertion seat tightly abuts against the sensor assembly so as to prevent the sensor assembly from being separated from a mounting portion of the base.

15. The insertion method as claimed in claim 11, wherein the first elastic member is configured as a pre-compressed spring, and has two opposite ends that respectively abut against an insertion seat and the cover body.

16. The insertion method as claimed in claim 11, wherein the second elastic member is configured as a pre-compressed spring, and has two opposite ends that respectively abut against an insertion seat and a retraction seat.

17. The insertion method as claimed in claim 11, wherein the insertion device further comprising a lower casing that is air-tightly and separably coupled to said cover body to form an airtight space within said insertion device, said airtight space being provided with a desiccant therein.
